BOQ TAKE-OFF

From architectural drawings to a checked Bill of Quantities

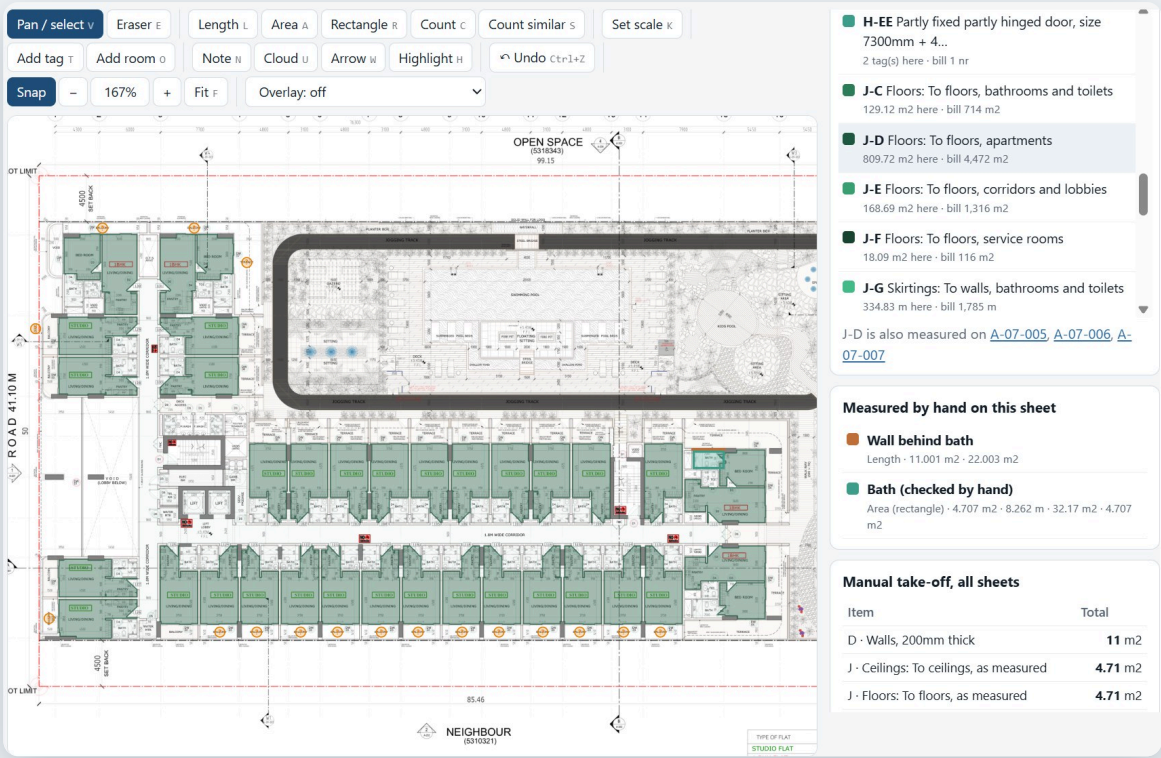
Reads a full drawing set, measures it by the POMI rules and writes the take-off and an unpriced bill in the house format. Every quantity can be traced to the drawing and corrected by hand.

42
drawings read in one run

~3 min
for the full set

61
bill items in 4 POMI sections

654
doors and windows, counted two ways



The Measure workspace: every apartment floor behind bill item J-D lit on the first floor plan.

Capability overview · September 2026
Figures and screenshots from the test set: Proposed Residential Building (B+G+5+Roof), plot 5310319, 42 architectural sheets.

AT A GLANCE

What goes in, what comes out

Give the platform a complete architectural set. It reads every sheet, finds each plan's true scale, measures walls, rooms, doors and windows, applies the POMI measurement rules and checks its own output before handing it over.

In

- A whole architectural set: plans, sections, schedules and details together
- PDF, DWG, DXF, JPG, PNG or TIFF, or a .zip of them
- A revision number, and (only for sheets not read before) an AI key for what is drawn as pictures

Out

- **Unpriced BOQ** in the house format, rates left blank
- **Take-off**: the dimension paper behind every quantity, with live formulas
- **Gaps list**: everything the drawings leave unclear
- **Marked-up drawings** in the trade colours
- **Revision changes** against the previous run

Test set result

SECTION	ITEMS	QUANTITIES
D Masonry	4	16,090 m²
G Thermal & moisture	4	3,296 m² · 214 m
H Doors & windows	34	654 nr
J Finishes	19	49,385 m² · 9,662 m

Checks passed. 28 door/window marks closed, 6 open. 64 gaps listed. 5 sheets left for an AI reading (no key given).

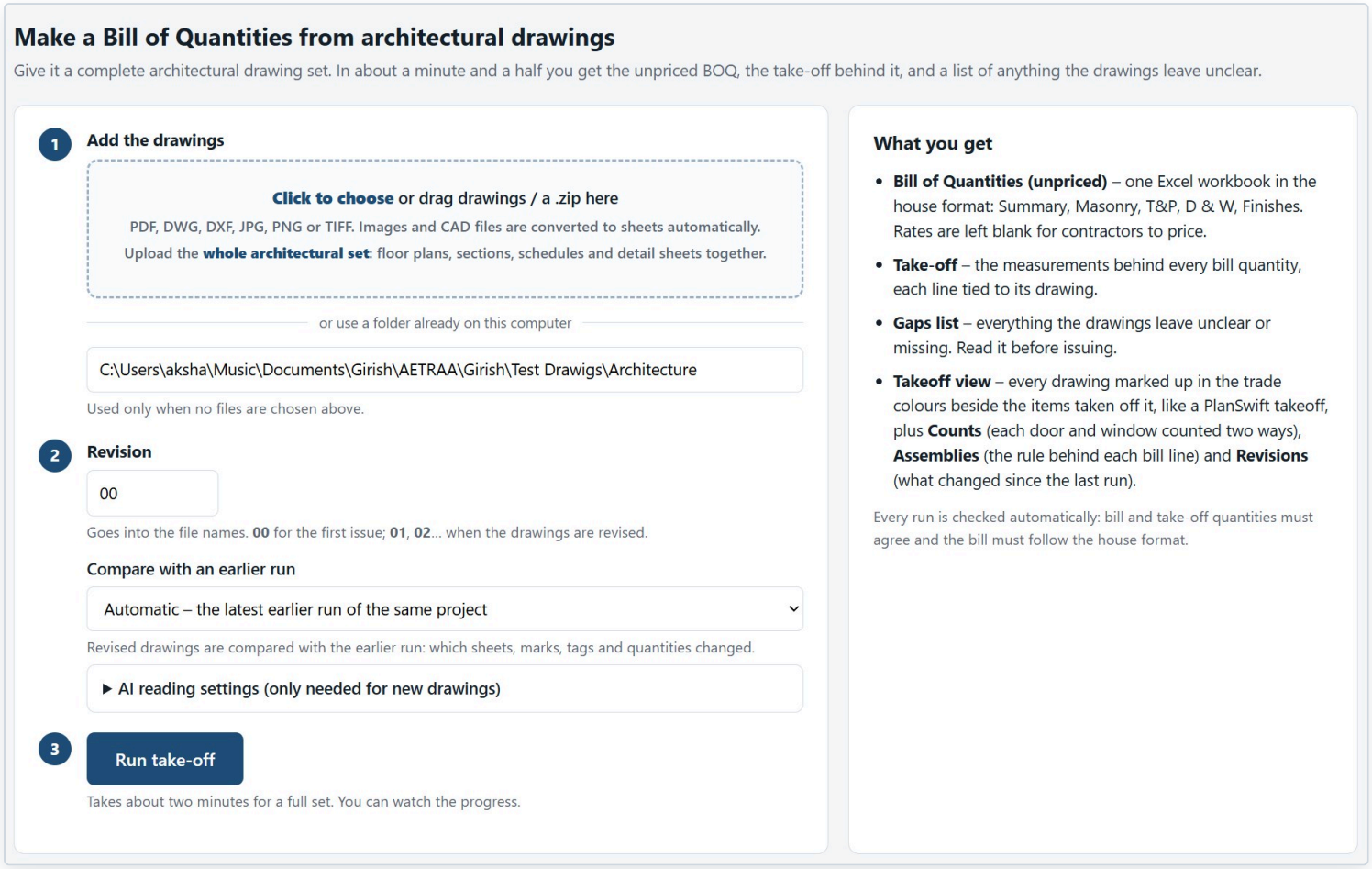
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RUNNING A TAKE-OFF

Upload the set, press Run

Upload the drawings or point to a folder on the computer, give the revision number and run. A full set of 42 sheets takes about three minutes, and the page shows each stage as it happens.



The start page, with the test drawing set selected.

INPUTS

- PDF, DWG, DXF, JPG, PNG, TIFF or a .zip; images and CAD files become sheets automatically
- The revision number goes into every file name; each run is compared with the last run of the project

AI READINGS

- Only what exists as graphics: schedule cells, levels, detail notes, outlined title blocks
- Claude first, then OpenAI, then Gemini. Keys are used for one run and not stored
- Readings are saved and reused, so unchanged drawings cost nothing to re-run

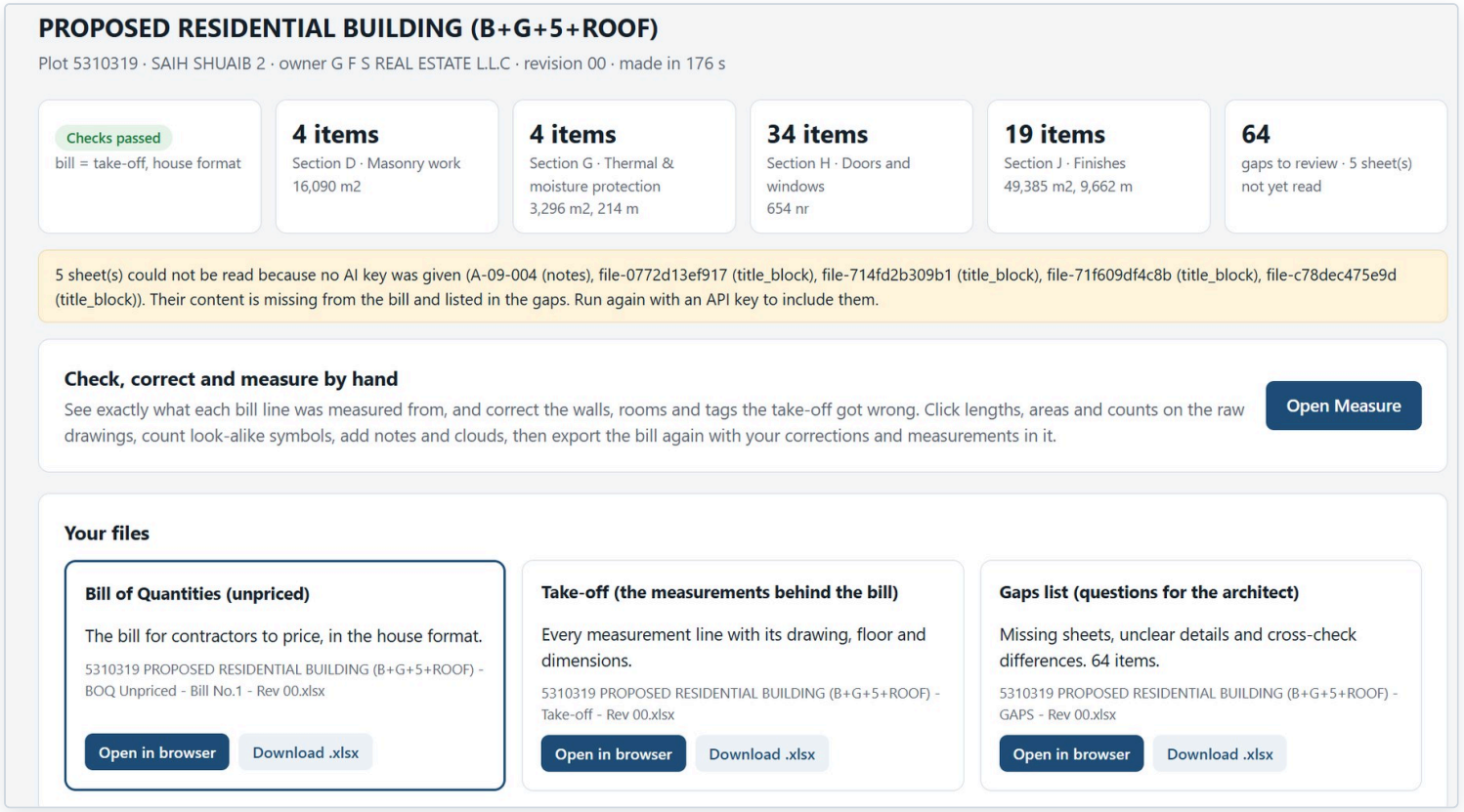


Live progress: title blocks, schedules, scale, measuring, POMI rules, workbooks, mark-ups, checks.

THE RESULTS

One page for the whole run

When the run finishes, a single page gives the check result, the items and quantities for each POMI section, the gaps to review and every file produced.



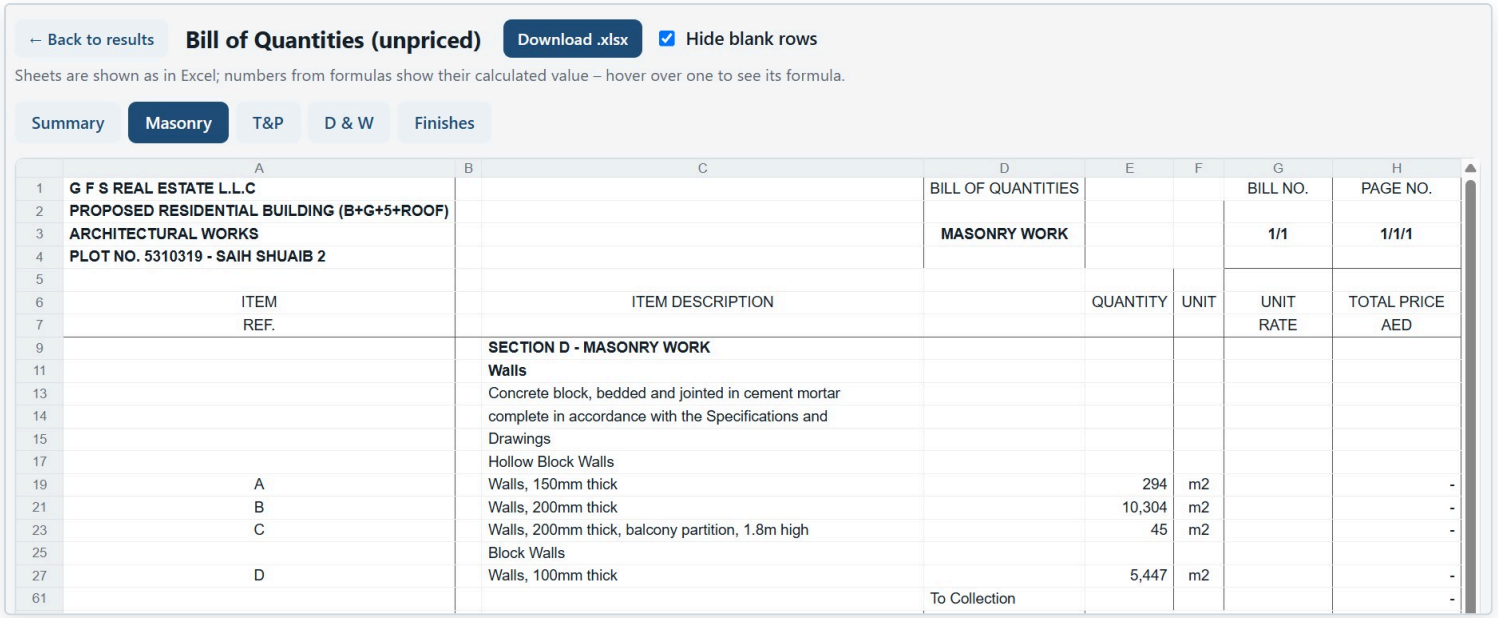
Results for the test set: made in 176 s, checks passed, 61 items, 64 gaps to review.

WHAT IT SHOWS

- **Checks passed**: every bill quantity equals its take-off, and the bill follows the house format
- Items and totals per section (D, G, H, J)
- Sheets that need an AI reading but had no key are named; their content goes to the gaps, never guessed
- Each workbook opens in the browser, with sheet tabs and calculated formulas, or downloads as .xlsx
- Every drawing read from, linked to its mark-up
- **Open Measure** to check, correct and measure by hand

An unpriced bill in the house format

One bill workbook, laid out as the issued sample bills: Summary, Masonry, Thermal & Protection, Doors & Windows and Finishes, with rates left blank for contractors to price.



	A	B	C	D	E	F	G	H
1	G F S REAL ESTATE L.L.C			BILL OF QUANTITIES			BILL NO.	PAGE NO.
2	PROPOSED RESIDENTIAL BUILDING (B+G+5+ROOF)							
3	ARCHITECTURAL WORKS			MASONRY WORK			1/1	1/1/1
4	PLOT NO. 5310319 - SAIH SHUAIB 2							
5								
6	ITEM		ITEM DESCRIPTION		QUANTITY	UNIT	UNIT	TOTAL PRICE
7	REF.						RATE	AED
9			SECTION D - MASONRY WORK					
11			Walls					
13			Concrete block, bedded and jointed in cement mortar					
14			complete in accordance with the Specifications and					
15			Drawings					
17			Hollow Block Walls					
19	A		Walls, 150mm thick		294	m2		-
21	B		Walls, 200mm thick		10,304	m2		-
23	C		Walls, 200mm thick, balcony partition, 1.8m high		45	m2		-
25			Block Walls					
27	D		Walls, 100mm thick		5,447	m2		-
61				To Collection				-

The Masonry sheet in the browser viewer (blank rows hidden).

LAYOUT, AS THE SAMPLES

- Band with client, project, package and plot on every page; bill and page numbers
- Headings, Note blocks and description paragraphs; short item lines; (Cont'd) headings
- 62-row pages with To Collection, then collection and summary pages
- Item letters restart on each page and skip I and O
- Fonts, column widths and page layout checked against the sample bills on every run

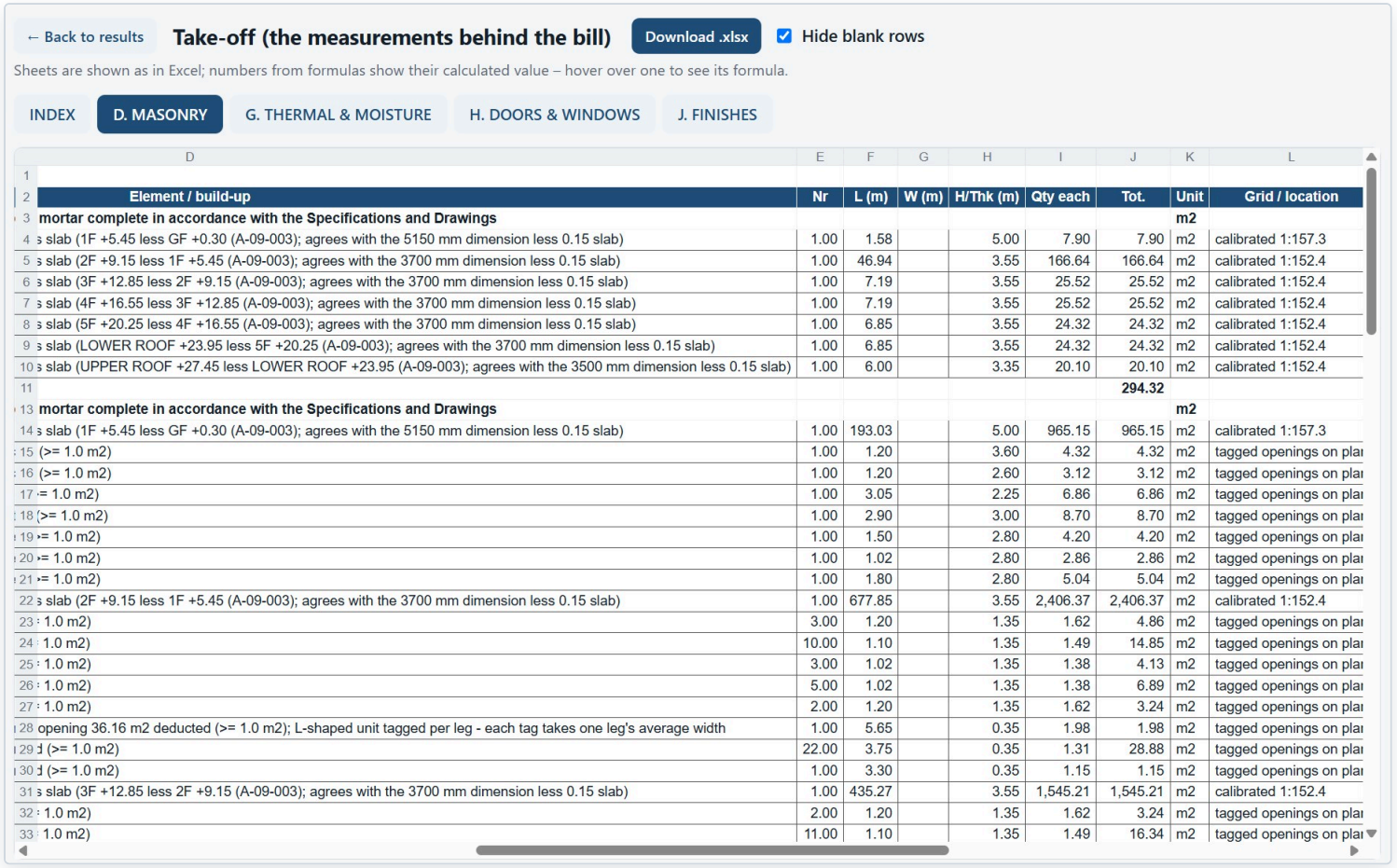
SHOWN

- Masonry: block walls by thickness, e.g. 200 mm walls 10,304 m²

THE TAKE-OFF

The measurements behind every bill quantity

Each bill quantity is built from take-off lines: drawing, floor, Nr × L × H, and how the figure was obtained. Quantities are live Excel formulas over the dimensions, not pasted numbers.



← Back to results **Take-off (the measurements behind the bill)** Download .xlsx ☑ Hide blank rows

Sheets are shown as in Excel; numbers from formulas show their calculated value – hover over one to see its formula.

INDEX | D. MASONRY | G. THERMAL & MOISTURE | H. DOORS & WINDOWS | J. FINISHES

Element / build-up	Nr	L (m)	W (m)	H/Thk (m)	Qty each	Tot.	Unit	Grid / location
mortar complete in accordance with the Specifications and Drawings							**m2**	
s slab (1F +5.45 less GF +0.30 (A-09-003); agrees with the 5150 mm dimension less 0.15 slab)	1.00	1.58		5.00	7.90	7.90	m2	calibrated 1:157.3
s slab (2F +9.15 less 1F +5.45 (A-09-003); agrees with the 3700 mm dimension less 0.15 slab)	1.00	46.94		3.55	166.64	166.64	m2	calibrated 1:152.4
s slab (3F +12.85 less 2F +9.15 (A-09-003); agrees with the 3700 mm dimension less 0.15 slab)	1.00	7.19		3.55	25.52	25.52	m2	calibrated 1:152.4
s slab (4F +16.55 less 3F +12.85 (A-09-003); agrees with the 3700 mm dimension less 0.15 slab)	1.00	7.19		3.55	25.52	25.52	m2	calibrated 1:152.4
s slab (5F +20.25 less 4F +16.55 (A-09-003); agrees with the 3700 mm dimension less 0.15 slab)	1.00	6.85		3.55	24.32	24.32	m2	calibrated 1:152.4
s slab (LOWER ROOF +23.95 less 5F +20.25 (A-09-003); agrees with the 3700 mm dimension less 0.15 slab)	1.00	6.85		3.55	24.32	24.32	m2	calibrated 1:152.4
s slab (UPPER ROOF +27.45 less LOWER ROOF +23.95 (A-09-003); agrees with the 3500 mm dimension less 0.15 slab)	1.00	6.00		3.35	20.10	20.10	m2	calibrated 1:152.4
						294.32		
mortar complete in accordance with the Specifications and Drawings							**m2**	
s slab (1F +5.45 less GF +0.30 (A-09-003); agrees with the 5150 mm dimension less 0.15 slab)	1.00	193.03		5.00	965.15	965.15	m2	calibrated 1:157.3
(>= 1.0 m2)	1.00	1.20		3.60	4.32	4.32	m2	tagged openings on pla
(>= 1.0 m2)	1.00	1.20		2.60	3.12	3.12	m2	tagged openings on pla
= 1.0 m2)	1.00	3.05		2.25	6.86	6.86	m2	tagged openings on pla
(>= 1.0 m2)	1.00	2.90		3.00	8.70	8.70	m2	tagged openings on pla
>= 1.0 m2)	1.00	1.50		2.80	4.20	4.20	m2	tagged openings on pla
>= 1.0 m2)	1.00	1.02		2.80	2.86	2.86	m2	tagged openings on pla
>= 1.0 m2)	1.00	1.80		2.80	5.04	5.04	m2	tagged openings on pla
s slab (2F +9.15 less 1F +5.45 (A-09-003); agrees with the 3700 mm dimension less 0.15 slab)	1.00	677.85		3.55	2,406.37	2,406.37	m2	calibrated 1:152.4
1.0 m2)	3.00	1.20		1.35	1.62	4.86	m2	tagged openings on pla
1.0 m2)	10.00	1.10		1.35	1.49	14.85	m2	tagged openings on pla
1.0 m2)	3.00	1.02		1.35	1.38	4.13	m2	tagged openings on pla
1.0 m2)	5.00	1.02		1.35	1.38	6.89	m2	tagged openings on pla
1.0 m2)	2.00	1.20		1.35	1.62	3.24	m2	tagged openings on pla
opening 36.16 m2 deducted (>= 1.0 m2); L-shaped unit tagged per leg - each tag takes one leg's average width	1.00	5.65		0.35	1.98	1.98	m2	tagged openings on pla
d (>= 1.0 m2)	22.00	3.75		0.35	1.31	28.88	m2	tagged openings on pla
d (>= 1.0 m2)	1.00	3.30		0.35	1.15	1.15	m2	tagged openings on pla
s slab (3F +12.85 less 2F +9.15 (A-09-003); agrees with the 3700 mm dimension less 0.15 slab)	1.00	435.27		3.55	1,545.21	1,545.21	m2	calibrated 1:152.4
1.0 m2)	2.00	1.20		1.35	1.62	3.24	m2	tagged openings on pla
1.0 m2)	11.00	1.10		1.35	1.49	16.34	m2	tagged openings on pla

Masonry dimension paper: 200 mm walls per floor, then the walling at each opening.

EVERY LINE SAYS

- Which drawing (a hyperlink to the sheet) and which floor
- The dimensions, and the scale it was measured at, e.g. *calibrated 1:152.4*
- Its origin: STATED, COUNTED, MEASURED, READ (VISUAL) or DERIVED; hand work as MEASURED (MANUAL) or (CORRECTED)
- Where heights came from: storey levels on the wall section, less the slab

POMI RULES APPLIED

- Walling over and under each tagged opening is added back
- Openings of 1 m² or more are deducted (GP3.2)

THE GAPS LIST

Nothing is assumed

Anything the drawings leave unclear or missing goes to the gaps workbook instead of into the bill: missing sheets, illegible dimensions, counts that disagree, rooms that could not be measured, work this version does not measure. The test set gave 64 gaps; ten are shown here as written.

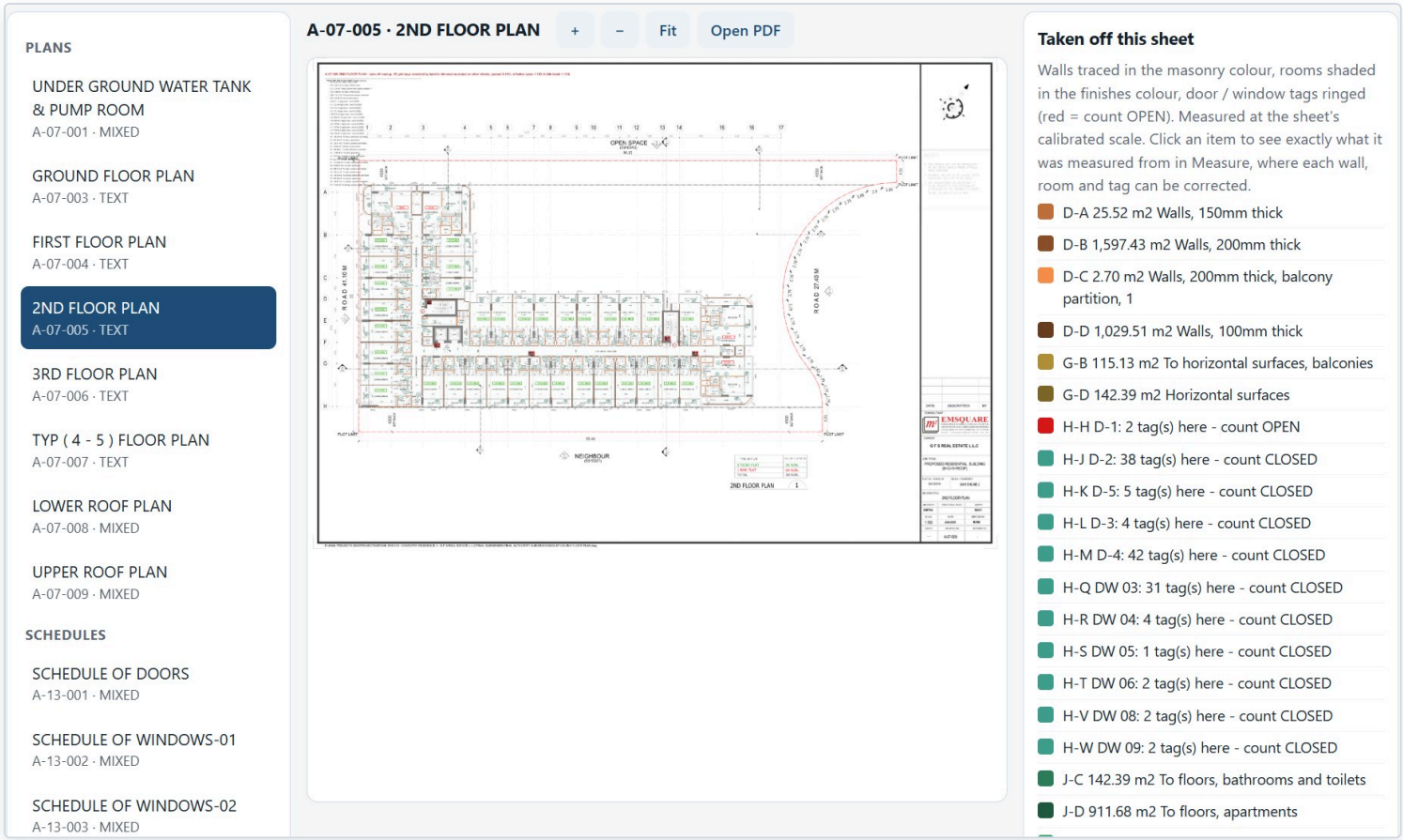
SEC.	SHEET	FIELD	REASON
(all)	A-07-002	sheet	'BASEMENT FLOOR PLAN' is on the list of drawings (A-01-001) but not in the set
(all)	A1\A-00-001 COVERSHEET.pdf	drawing_no	no drawing number printed on this sheet (read visually); sheet not used - a file name is never taken as a drawing number
(all)	A-07-003	grid G-H	not legible; taken as 7900 mm = overall 39500 less the other bays (DERIVED)
(all)	A-09-004	notes reading	content is drawn as graphics and was not read (no Anthropic credentials; no OpenAI credentials; no Gemini credentials); request left in readings/A-09-004.notes.request/
D	(set)	slab thickness	150 mm taken from '15cm CONCRETE SLAB' (TYPICAL FLOOR FINISH DETAILS, A-12-012) (STATED) for wall heights
H	A-13-001	D-1 count	OPEN - schedule A-13-001 states 20, plans show 19 (A-07-001 1, A-07-003 4, A-07-004 3, A-07-005 2, A-07-006 2, A-07-007 4, A-07-008 2, A-07-009 1). Both kept; neither is billed until resolve…
J	A-07-005	room	BATH / BED ROOM: a wet and a dry room fill as one space (open doorway) - not measured
E	A-09-003	railings	1200mm laminated toughened glass railings and aluminium railings are noted; railing lengths are not measured by this version
J	A-08-001, A-08-002, A-08-003, A-08-004	external wall finishes	finishes shown in colour on the elevations are measured by area; surfaces left uncoloured (paint, aluminium profiles and slats) are not, and faces seen at an angle are taken as drawn (projec…
J	A-08-004	finish F1	87.3 m2 coloured #E6C1BE: no callout names it - not billed

Excerpt from the GAPS workbook of the test set. A gap closes by itself when a correction resolves it, for example a room drawn by hand round a label the take-off could not measure (page 17).

MARKED-UP DRAWINGS

The take-off drawn on every sheet

Every drawing the take-off read from is marked up in the house trade colours and shown beside the items taken off it, like a PlanSwift takeoff. The same mark-ups download as PDFs.



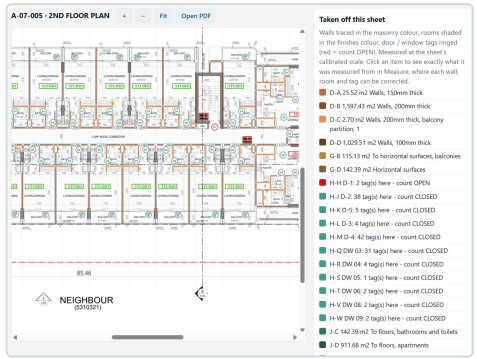
Takeoff view of the 2nd floor plan with its items.

ON EACH PLAN

- Walls traced in the masonry colour, one shade per thickness
- Rooms shaded in the finishes colour
- Door and window tags ringed; red when the count is open
- A legend of every item on the sheet with its quantity from that sheet
- Each legend item links to Measure, where its shapes light up

ON EACH SCHEDULE

- The regions read off the image outlined, and each mark with its count and status

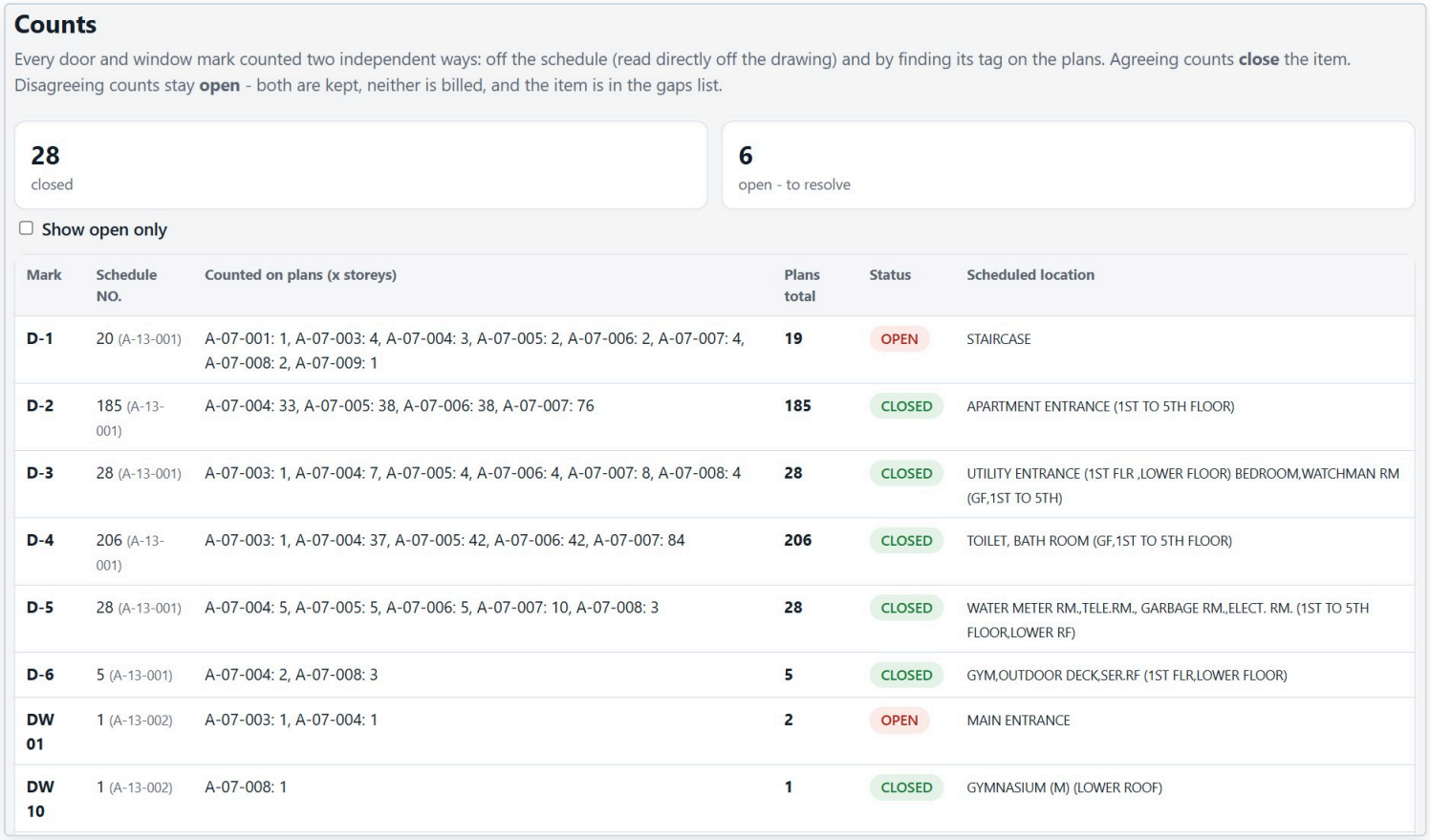


Close-up: walls, rooms and tags in the trade colours.

COUNTS

Every door and window counted two ways

Each mark is counted twice, independently: from the schedule's NO. cell and from its tags on the plans, times the storeys a typical plan serves. Only agreeing counts are billed.



Counts

Every door and window mark counted two independent ways: off the schedule (read directly off the drawing) and by finding its tag on the plans. Agreeing counts **close** the item. Disagreeing counts stay **open** - both are kept, neither is billed, and the item is in the gaps list.

28 closed

6 open - to resolve

☐ Show open only

Mark	Schedule NO.	Counted on plans (x storeys)	Plans total	Status	Scheduled location
D-1	20 (A-13-001)	A-07-001: 1, A-07-003: 4, A-07-004: 3, A-07-005: 2, A-07-006: 2, A-07-007: 4, A-07-008: 2, A-07-009: 1	**19**	OPEN	STAIRCASE
D-2	185 (A-13-001)	A-07-004: 33, A-07-005: 38, A-07-006: 38, A-07-007: 76	**185**	CLOSED	APARTMENT ENTRANCE (1ST TO 5TH FLOOR)
D-3	28 (A-13-001)	A-07-003: 1, A-07-004: 7, A-07-005: 4, A-07-006: 4, A-07-007: 8, A-07-008: 4	**28**	CLOSED	UTILITY ENTRANCE (1ST FLR ,LOWER FLOOR) BEDROOM,WATCHMAN RM (GF,1ST TO 5TH)
D-4	206 (A-13-001)	A-07-003: 1, A-07-004: 37, A-07-005: 42, A-07-006: 42, A-07-007: 84	**206**	CLOSED	TOILET, BATH ROOM (GF,1ST TO 5TH FLOOR)
D-5	28 (A-13-001)	A-07-004: 5, A-07-005: 5, A-07-006: 5, A-07-007: 10, A-07-008: 3	**28**	CLOSED	WATER METER RM.,TELE.RM., GARBAGE RM.,ELECT. RM. (1ST TO 5TH FLOOR,LOWER RF)
D-6	5 (A-13-001)	A-07-004: 2, A-07-008: 3	**5**	CLOSED	GYM,OUTDOOR DECK,SER.RF (1ST FLR,LOWER FLOOR)
DW 01	1 (A-13-002)	A-07-003: 1, A-07-004: 1	**2**	OPEN	MAIN ENTRANCE
DW 10	1 (A-13-002)	A-07-008: 1	**1**	CLOSED	GYMNASIUM (M) (LOWER ROOF)

The Counts tab: schedule number, tags found on each plan, and status.

RULES

- Agreeing counts CLOSED close the item
- Disagreeing counts stay OPEN: both go to the gaps and the bill quantity is left blank
- An L-shaped unit tagged on each leg counts once
- Tags on floors the schedule does not name are reported, not counted

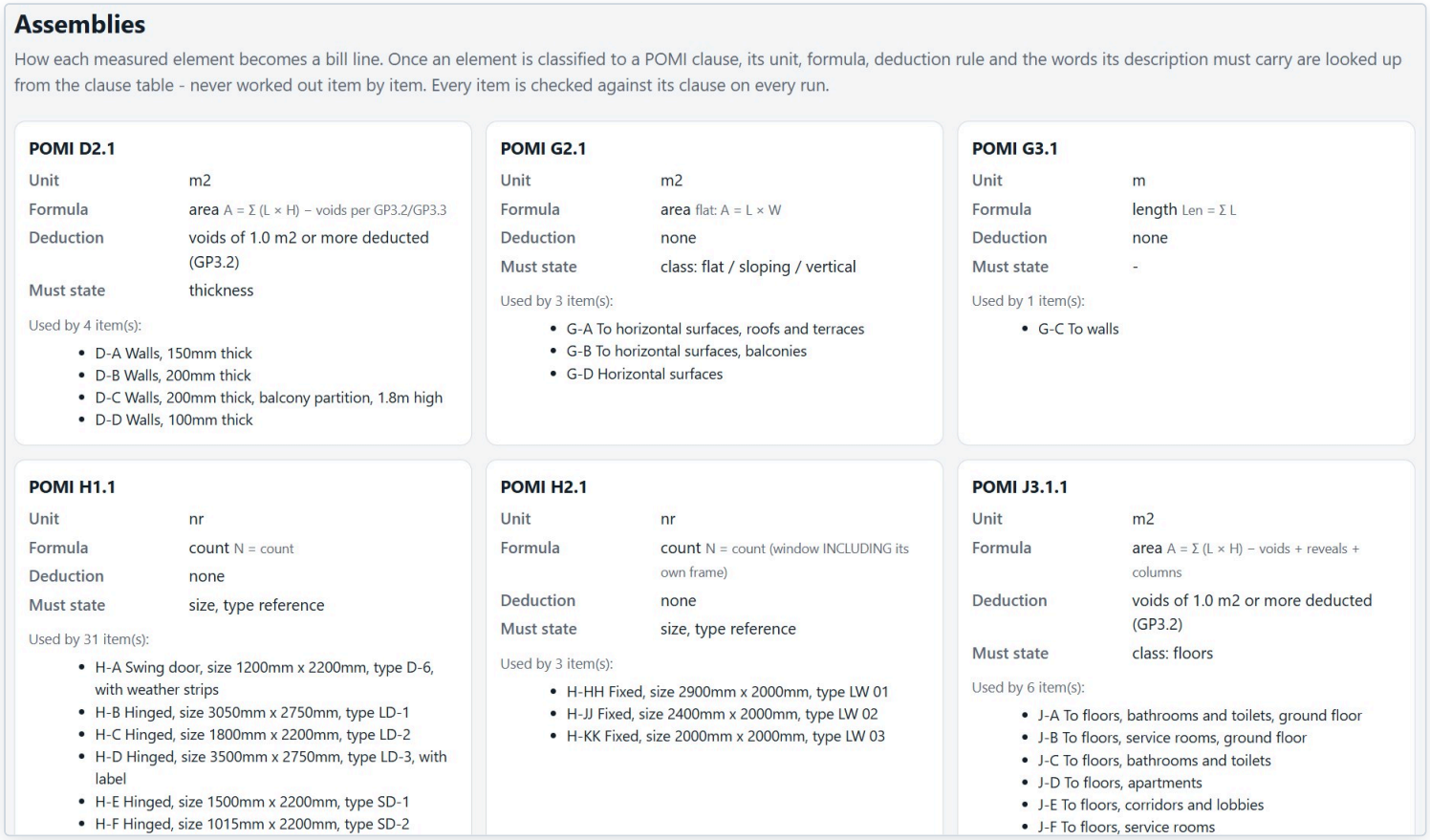
TEST SET

- 28 marks closed, 6 open, e.g. D-1: schedule 20, plans 19
- LW 03 left open: scheduled in the basement, which has no plan in the set

CLAUSE RULES

Each bill line follows its POMI clause

Once an element is classified to a POMI clause, its unit, formula, deduction rule and the words its description must carry are looked up from the clause table, and every item is checked against its clause on every run.



Assemblies

How each measured element becomes a bill line. Once an element is classified to a POMI clause, its unit, formula, deduction rule and the words its description must carry are looked up from the clause table - never worked out item by item. Every item is checked against its clause on every run.

POMI D2.1

Unit	m2
Formula	area A = Σ (L × H) − voids per GP3.2/GP3.3
Deduction	voids of 1.0 m2 or more deducted (GP3.2)
Must state	thickness

Used by 4 item(s):

- D-A Walls, 150mm thick
- D-B Walls, 200mm thick
- D-C Walls, 200mm thick, balcony partition, 1.8m high
- D-D Walls, 100mm thick

POMI G2.1

Unit	m2
Formula	area flat: A = L × W
Deduction	none
Must state	class: flat / sloping / vertical

Used by 3 item(s):

- G-A To horizontal surfaces, roofs and terraces
- G-B To horizontal surfaces, balconies
- G-D Horizontal surfaces

POMI G3.1

Unit	m
Formula	length Len = Σ L
Deduction	none
Must state	-

Used by 1 item(s):

- G-C To walls

POMI H1.1

Unit	nr
Formula	count N = count
Deduction	none
Must state	size, type reference

Used by 31 item(s):

- H-A Swing door, size 1200mm x 2200mm, type D-6, with weather strips
- H-B Hinged, size 3050mm x 2750mm, type LD-1
- H-C Hinged, size 1800mm x 2200mm, type LD-2
- H-D Hinged, size 3500mm x 2750mm, type LD-3, with label
- H-E Hinged, size 1500mm x 2200mm, type SD-1
- H-F Hinged, size 1015mm x 2200mm, type SD-2

POMI H2.1

Unit	nr
Formula	count N = count (window INCLUDING its own frame)
Deduction	none
Must state	size, type reference

Used by 3 item(s):

- H-HH Fixed, size 2900mm x 2000mm, type LW 01
- H-JJ Fixed, size 2400mm x 2000mm, type LW 02
- H-KK Fixed, size 2000mm x 2000mm, type LW 03

POMI J3.1.1

Unit	m2
Formula	area A = Σ (L × H) − voids + reveals + columns
Deduction	voids of 1.0 m2 or more deducted (GP3.2)
Must state	class: floors

Used by 6 item(s):

- J-A To floors, bathrooms and toilets, ground floor
- J-B To floors, service rooms, ground floor
- J-C To floors, bathrooms and toilets
- J-D To floors, apartments
- J-E To floors, corridors and lobbies
- J-F To floors, service rooms

The Assemblies tab: each clause used, its rule, and the items it governs.

EXAMPLES

- **D2.1** walls: m^2, area less voids of 1 m^2 or more; must state thickness
- **H1.1** doors: enumerated; must state size and type reference
- **G2.1** waterproofing: m^2; must state flat, sloping or vertical
- **J3.1** finishes: m^2 by class (floors, walls, ceilings)

CHECKED

- Unit, the shape of the dimension lines and the required wording. An item that fails is written to the gaps

SHEETS

Every drawing registered from its own title block

Each sheet is identified from its title block, never its file name, and classified by how it can be read. Content drawn as graphics is read off the image instead of being skipped.

Sheets

Every drawing in the set and how it could be read. Where the text layer is empty or garbled - outlined CAD text, scans - the region is read directly off the image instead of being skipped. A DXF placed beside a PDF (same name) is measured from its native geometry.

Drawing	Title	Kind	Readability	How it was read	Title block	File
A-01-001	LIST OF DRAWINGS	general	TEXT	text layer	STATED	A-01-001 LIST OF DRAWINGS.pdf
G-102	PERSPECTIVES	general	MIXED	text layer + direct visual reading where text is outlined	STATED	A-02-001 PERSPECTIVES-G-102.pdf
A-03-001	AREA CALCULATION-1	area	TEXT	text layer	READ (VISUAL)	A-03-001.pdf
A-03-004	AREA CALCULATION-4	area	MIXED	text layer + direct visual reading where text is outlined	READ (VISUAL)	A-03-001TOA-03-004 AREA CALCULATION-A-03-004.pdf
A-03-002	AREA CALCULATION-2	area	TEXT	text layer	STATED	A-03-002 AREACALCULATION-2.pdf
A-03-003	AREA CALCULATION-3	area	MIXED	text layer + direct visual reading where text is outlined	STATED	A-03-003 22NOS AREACALCULATION-3.pdf
A-06-001	SETTING OUT PLAN	general	MIXED	text layer + direct visual reading where text is outlined	READ (VISUAL)	A-06-001 SETTINGOUT LAYOUT-A-100 SETTING OUT PLAN.pdf
A-07-001	UNDER GROUND WATER TANK & PUMP ROOM	plan	MIXED	text layer + direct visual reading where text is outlined	STATED	A-07-001 UNDER GROUND WATER TANK & PUMP ROOM.pdf
A-07-003	GROUND FLOOR PLAN	plan	TEXT	text layer	STATED	A-07-003 GROUND FLOOR PLAN.pdf
A-07-004	FIRST FLOOR PLAN	plan	TEXT	text layer	STATED	A-07-004 FIRST FLOOR PLAN.pdf

The Sheets tab: drawing, kind, readability and how each was read.

READING

- TEXT, OUTLINED, RASTER, MIXED or CAD, per page and region
- Outlined or scanned regions (schedules, title blocks, dimension strings) are cropped tightly and read visually
- A garbled value, such as a drawing number, is re-read visually
- A DXF beside a sheet is measured from its native geometry

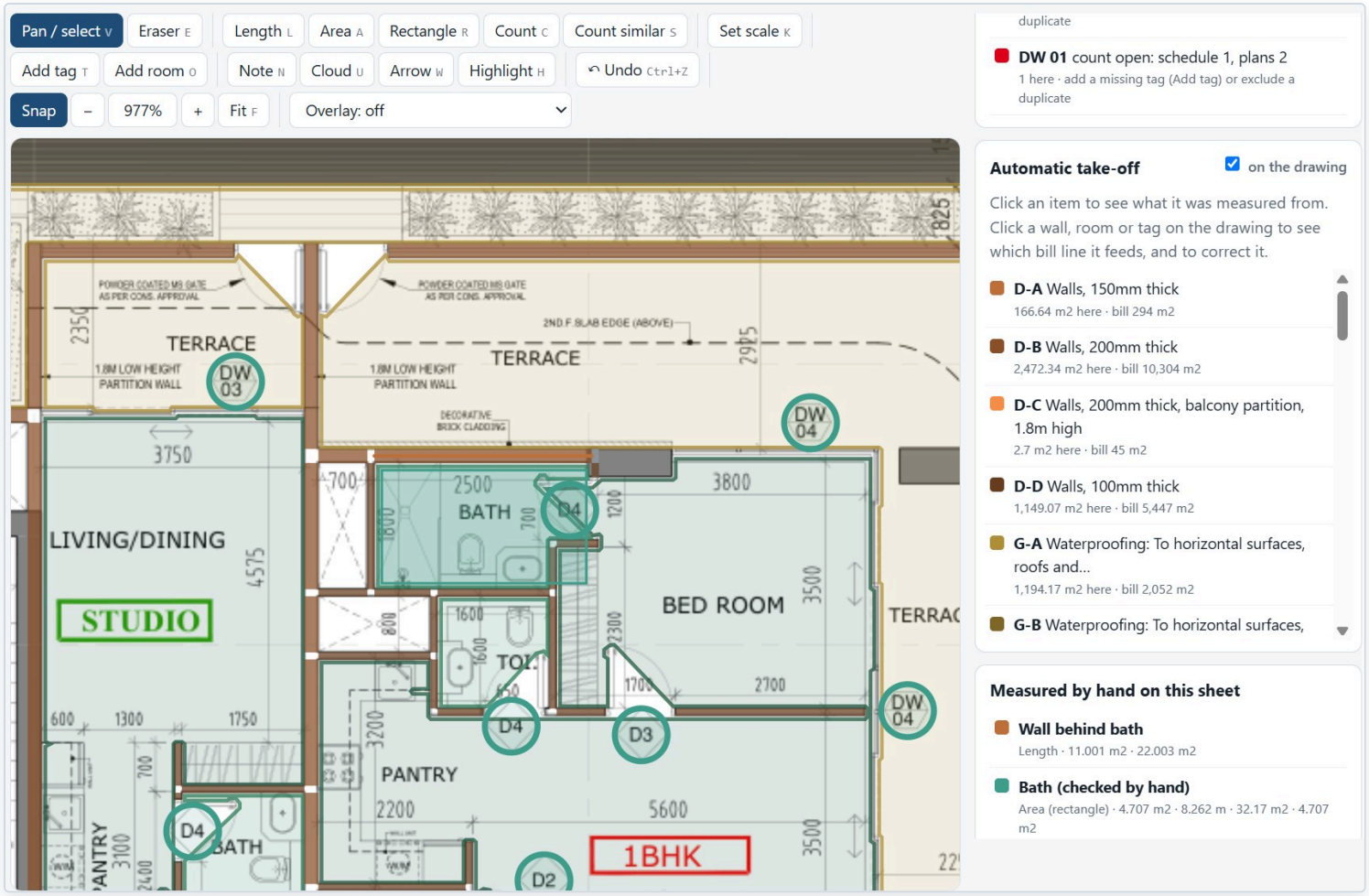
SCALE

- Never taken from the title block. Each plan is calibrated from its grid dimension chain
- First floor plan: title block says 1:150, measured 1:152.4, with 23 grid bays agreeing within 0.19%

TRACING

The take-off drawn over the plan, shape by shape

In Measure, every wall piece, room and tag the take-off measured is drawn over the plan in its trade colour. It is the same geometry the bill was calculated from, so nothing is hidden behind a number.



First floor plan, close up. A wall and a bath measured by hand appear too, in their own colours.

ON THE DRAWING

- Walls at their drawn thickness, coloured by the bill item they feed
- Wall pieces of no block size in grey dashes: not billed until given a thickness
- Rooms tinted by finish class; terraces and balconies in the waterproofing colour
- Tags ringed, red while the count is open

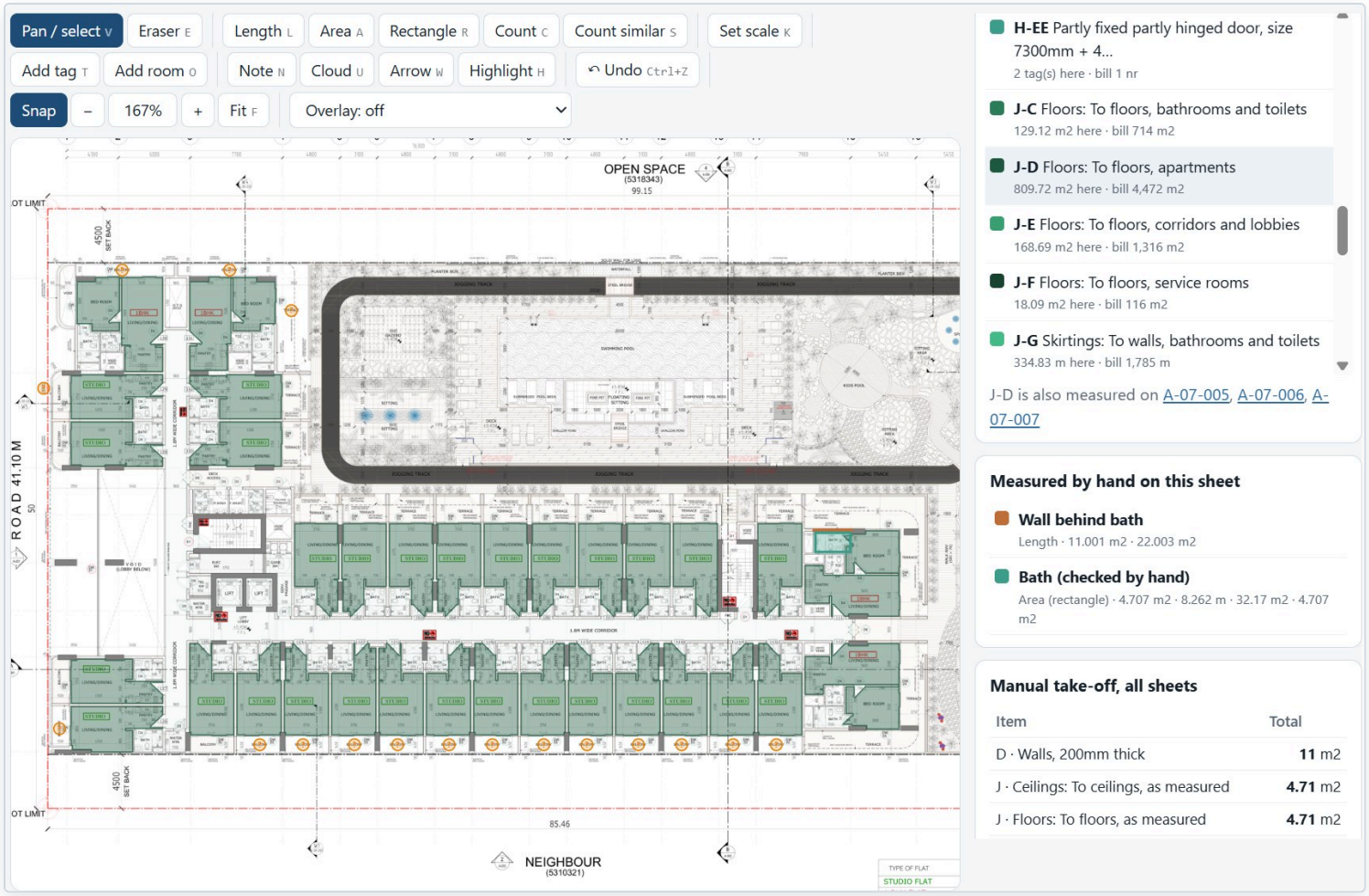
BESIDE IT

- Every item on the sheet with its quantity here and in the bill
- *To check on this sheet*: rooms not measured and open counts

TRACING

Follow any bill item to its shapes

Click an item, in the list here or in the Takeoff view legend, and its shapes light up while everything else fades. The view moves to them and names the other sheets the item is measured on.



Every apartment floor behind J-D lit on the first floor plan.

EXAMPLE

- **J-D To floors, apartments**: 809.72 m² on this plan, 4,472 m² in the bill
- Also measured on A-07-005, A-07-006 and A-07-007, one click away

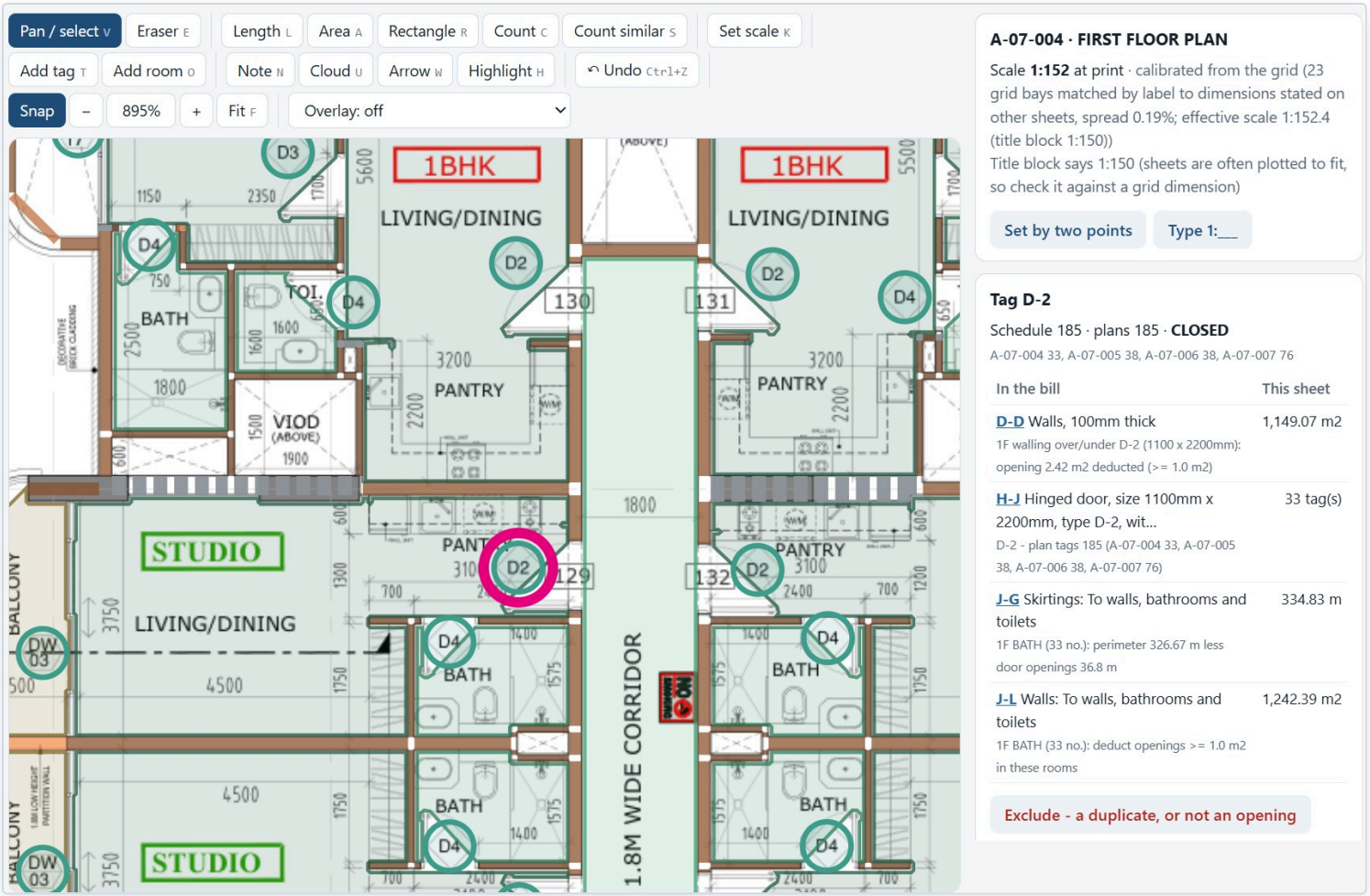
WHY IT MATTERS

- An estimator can check an automated quantity against the drawing in seconds
- Anything wrongly included or missed is visible at once

TRACING

Ask any shape which bill lines it feeds

Point at a wall, room or tag to see what it is and which items it feeds; click it for the take-off lines it is in. A single door tag, for example, feeds four items.



A D-2 tag selected on the first floor plan, with each line it feeds.

ONE D-2 DOOR TAG

- **H-J** the door count: 185 nr, schedule and plans agree
- **D-D** the walling over the door, added back to the 100 mm walls
- **J-G** the door width, taken off the skirting
- **J-L** the opening, deducted from the wall plaster

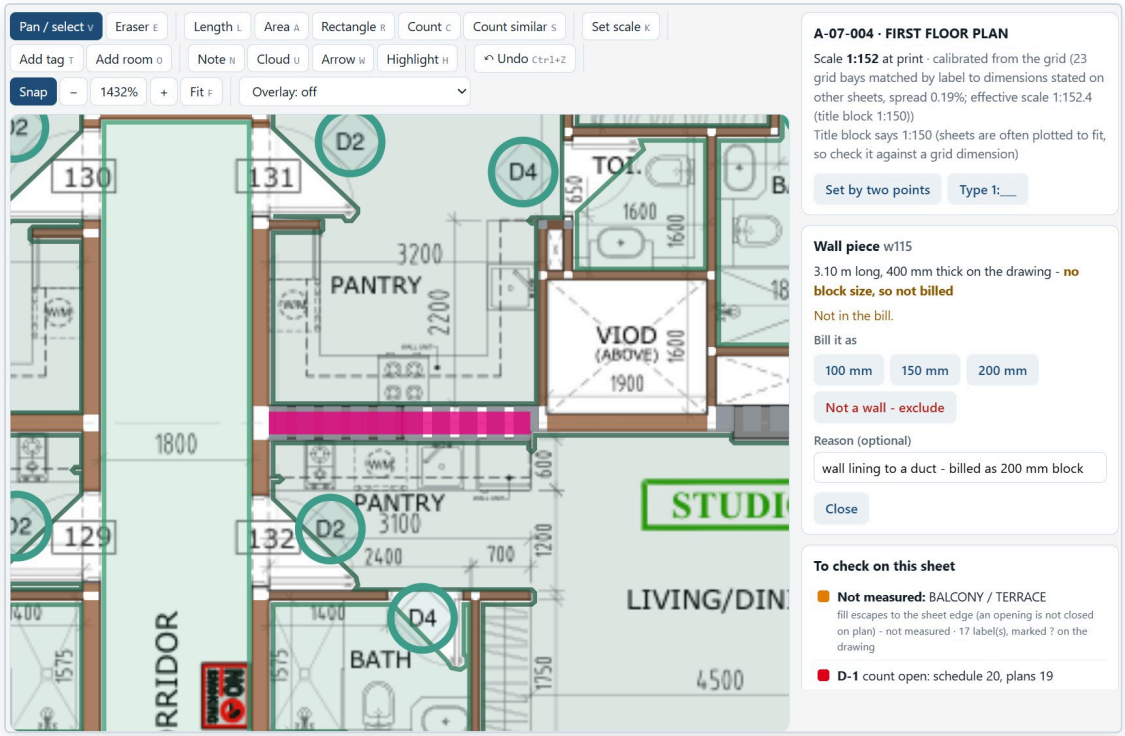
FOR A WALL

- Its length, the thickness drawn and the thickness billed, and every line it is in

CORRECTING

Fix the take-off where you see it

When the take-off got something wrong, correct it on the drawing. Each correction is measured again by the same rules straight away, so a door added also comes off the skirting and the plaster. With no correction, the automatic take-off is reproduced exactly.



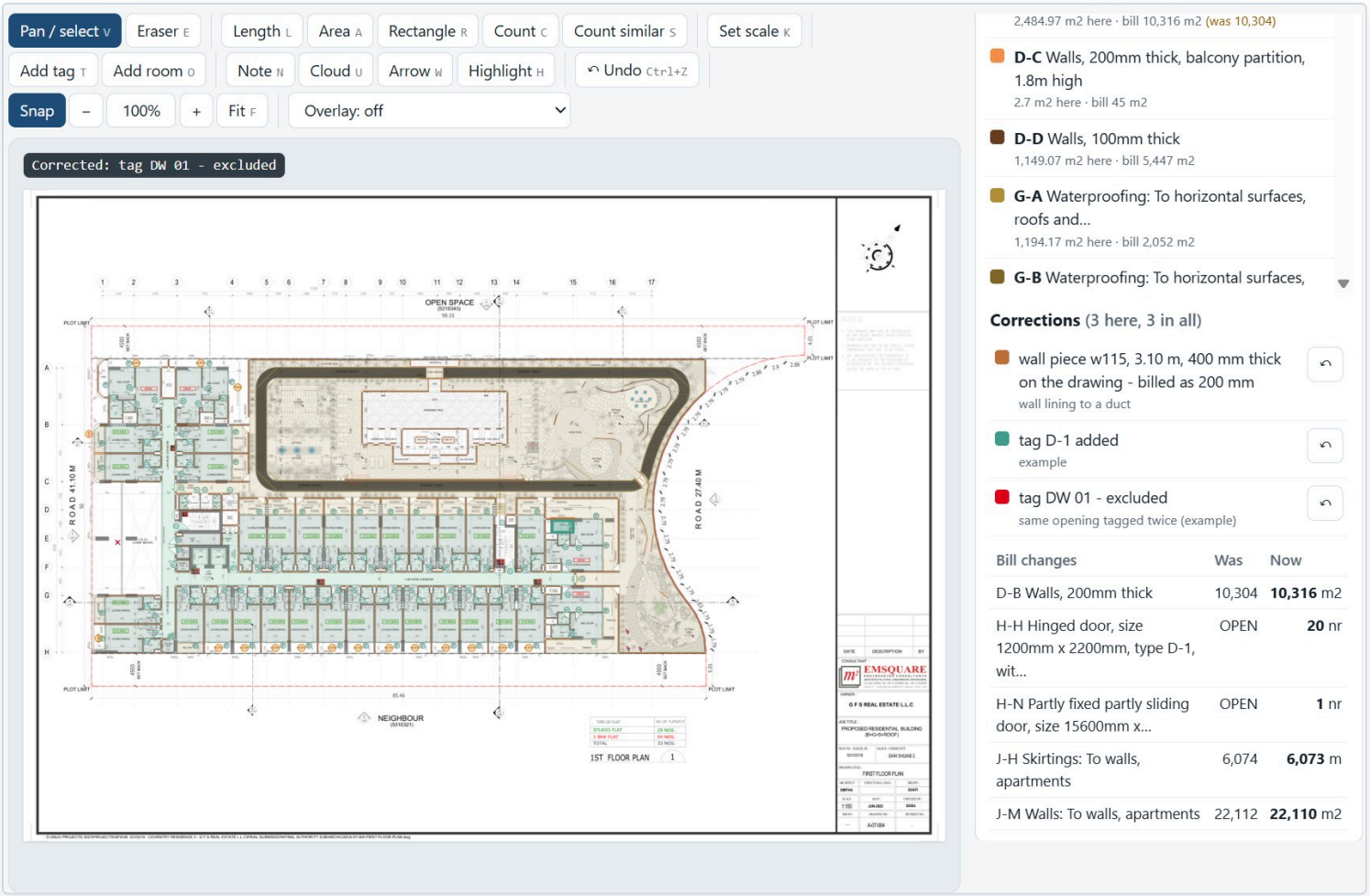
A wall piece drawn 400 mm thick, not a block size and so not billed, selected with its correction options.

CORRECTION	HOW	EFFECT
Wall thickness	Click a wall piece, choose a block size	Billed on its own line, MEASURED (CORRECTED)
Not a wall	Click it, *Not a wall - exclude* (or Delete)	Leaves its items; each item notes it
Rename a room	Click a room, choose its name	Moves to that room's finish class (and wet-area waterproofing if wet)
Not a room	Click it, *Not a room - exclude*	Leaves floors, walls, skirtings and ceilings
Duplicate tag	Click a tag, *Exclude*	Count reconciled again; the item closes if the counts now agree
Missing tag	**Add tag**: click where it is, choose the mark	Counted; its opening deducted and walling added back
Room not measured	**Add room** or click a ?: draw round it, name it	Billed MEASURED (MANUAL) with the room's finishes; the gap closes
Sloping room	Click a room, give its slope (15, 1:12, 8%)	Floor finish and waterproofing measured on the slope as sloping items
Name a colour	On an elevation, click a coloured area, say what it is	Billed as an external finish, MEASURED (CORRECTED); or *exclude* it

CORRECTING

Every correction listed, undoable, with its effect on the bill

Corrections are listed with their reasons and can be undone one by one (or with Ctrl+Z). The bill changes they caused are shown as was → now, and an excluded shape stays drawn in faint red so it can be put back.



Corrections and bill changes on the first floor plan.

EXAMPLE CORRECTIONS

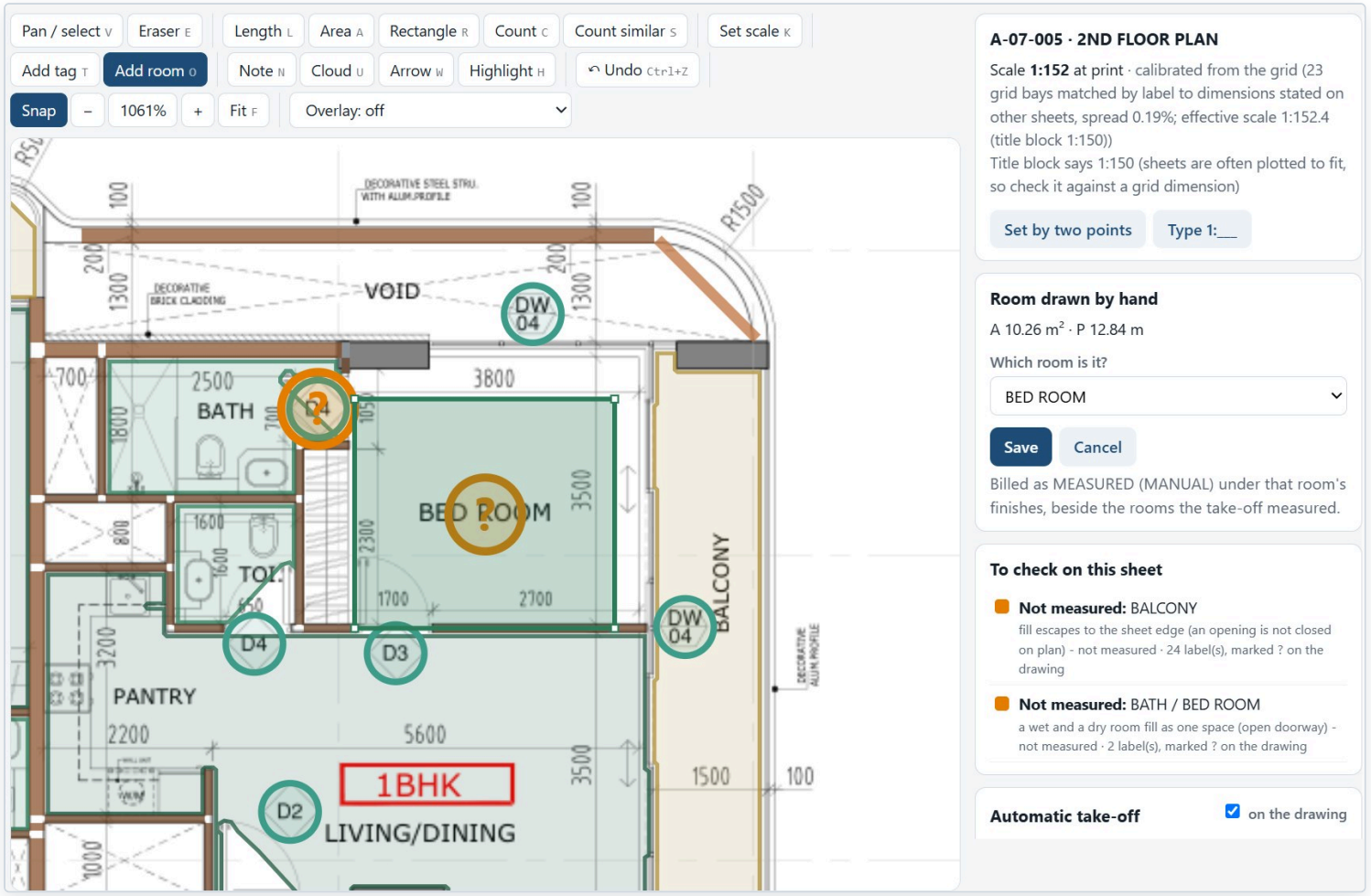
- A 3.10 m wall piece billed as 200 mm: **D-B 10,304 → 10,316 m²**
- A D-1 tag added: **H-H OPEN → 20 nr**. The rules also took the new door off the skirting and plaster: J-H 6,074 → 6,073 m, J-M 22,112 → 22,110 m²
- A duplicate DW 01 tag excluded: **H-N OPEN → 1 nr**

These corrections were made for this document to show the mechanism and then removed; they are not findings about the drawings.

CORRECTING

Draw the rooms the take-off could not measure

Where the flood fill could not measure a room, such as an open doorway or a fill that leaked outside, each label is marked **?** on the drawing and listed under *To check*. Draw round the room and name it.



2nd floor plan: the bath and bedroom labels marked ?, a bedroom drawn round by hand.

WHAT HAPPENS

- Billed as MEASURED (MANUAL) under that room's finishes: floor, skirting, wall plaster, ceiling; waterproofing if wet
- The gap closes once every label it covers has a room drawn round it
- Snapping to line ends and edge scrolling help trace the walls

EXAMPLE

- A bedroom the fill merged with its bath: 10.26 m², perimeter 12.84 m

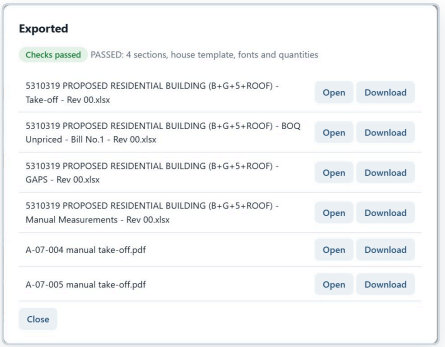
CORRECTING

The audit trail stays in the workbook

Export re-issues the take-off, bill and gaps with the corrections made and the manual measurements added, and runs the house-format check again. Every hand change sits on its own take-off line with its origin.

ITEM	DRAWING	FLOOR	TAKE-OFF LINE	QTY	ORIGIN
D-B	A-07-004	1F	200mm walls given their thickness by hand (1 piece(s)), net length on plan (openings excluded) x storey height less slab…	11.00	MEASURED (CORRECTED)
H-N	A-07-004		Corrected by hand on A-07-004: 1 tag(s) excluded (DW 01) - same opening tagged twice (example)		MEASURED (CORRECTED)
J-D	A-07-005	2F	BED ROOM (1 no.) - drawn by hand	10.26	MEASURED (MANUAL)
J-H	A-07-005	2F	BED ROOM (1 no.) - drawn by hand: perimeter 12.84 m less door openings 1.02 m	11.83	MEASURED (MANUAL)
J-M	A-07-005	2F	BED ROOM (1 no.) - drawn by hand: room perimeter x storey height less slab (3F +12.85 less 2F +9.15 (A-09-003); agrees w…	45.58	MEASURED (MANUAL)
J-R	A-07-005	2F	BED ROOM (1 no.) - drawn by hand	10.26	MEASURED (MANUAL)

Lines from the exported take-off after the example corrections (quantities in the item's unit).

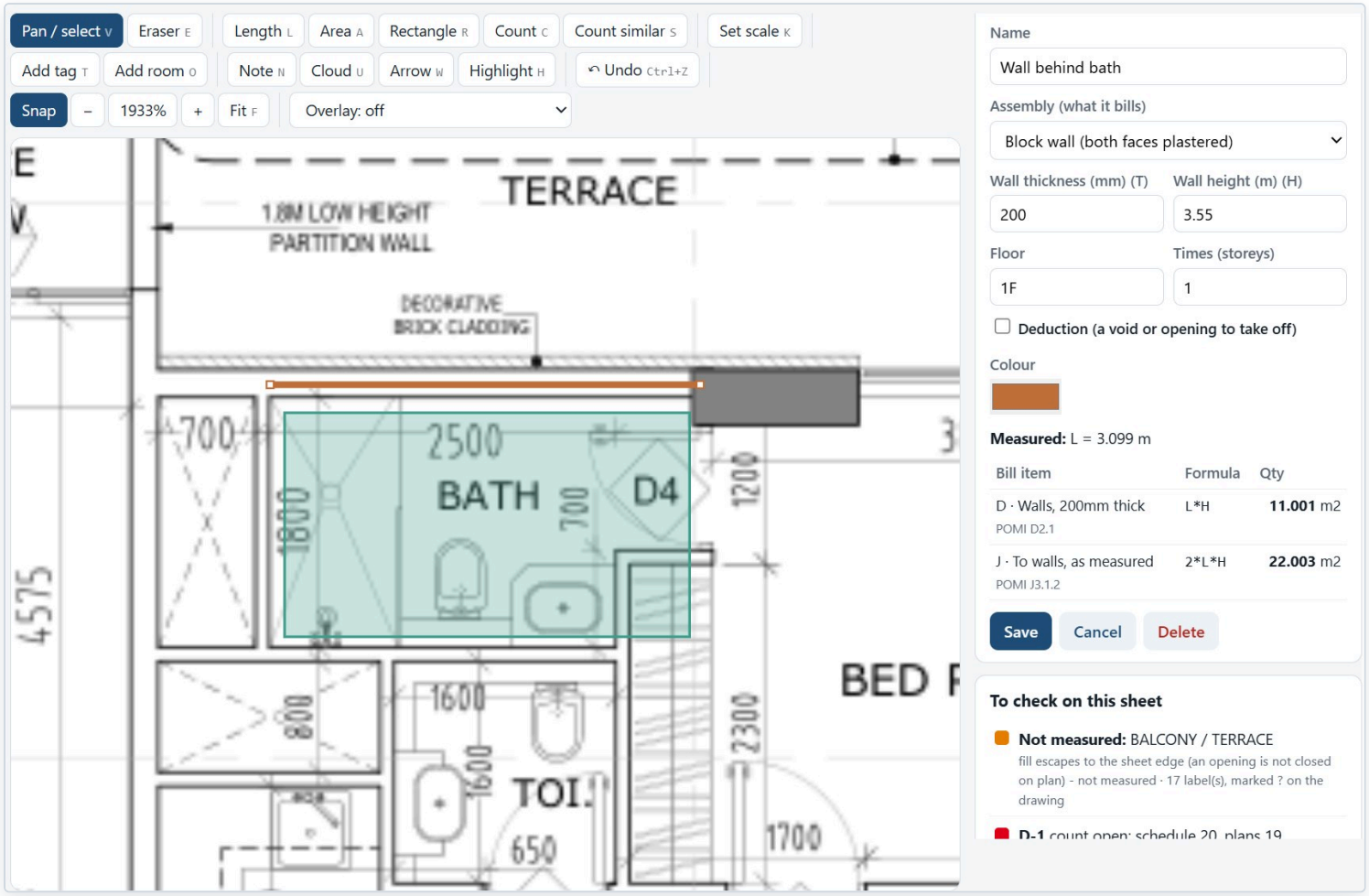


Export: checks passed; take-off, bill, gaps, a Manual Measurements workbook with a CORRECTIONS tab, and marked-up PDFs.

MEASURING BY HAND

PlanSwift-style tools on the raw drawing

Length, area, rectangle and count, at the plan's calibrated scale (or one set from two points), snapping to the ends of drawn lines. An assembly turns one measurement into all the bill lines it needs.



A wall measured by hand on the first floor plan, with the two bill lines its assembly makes.

EXAMPLE

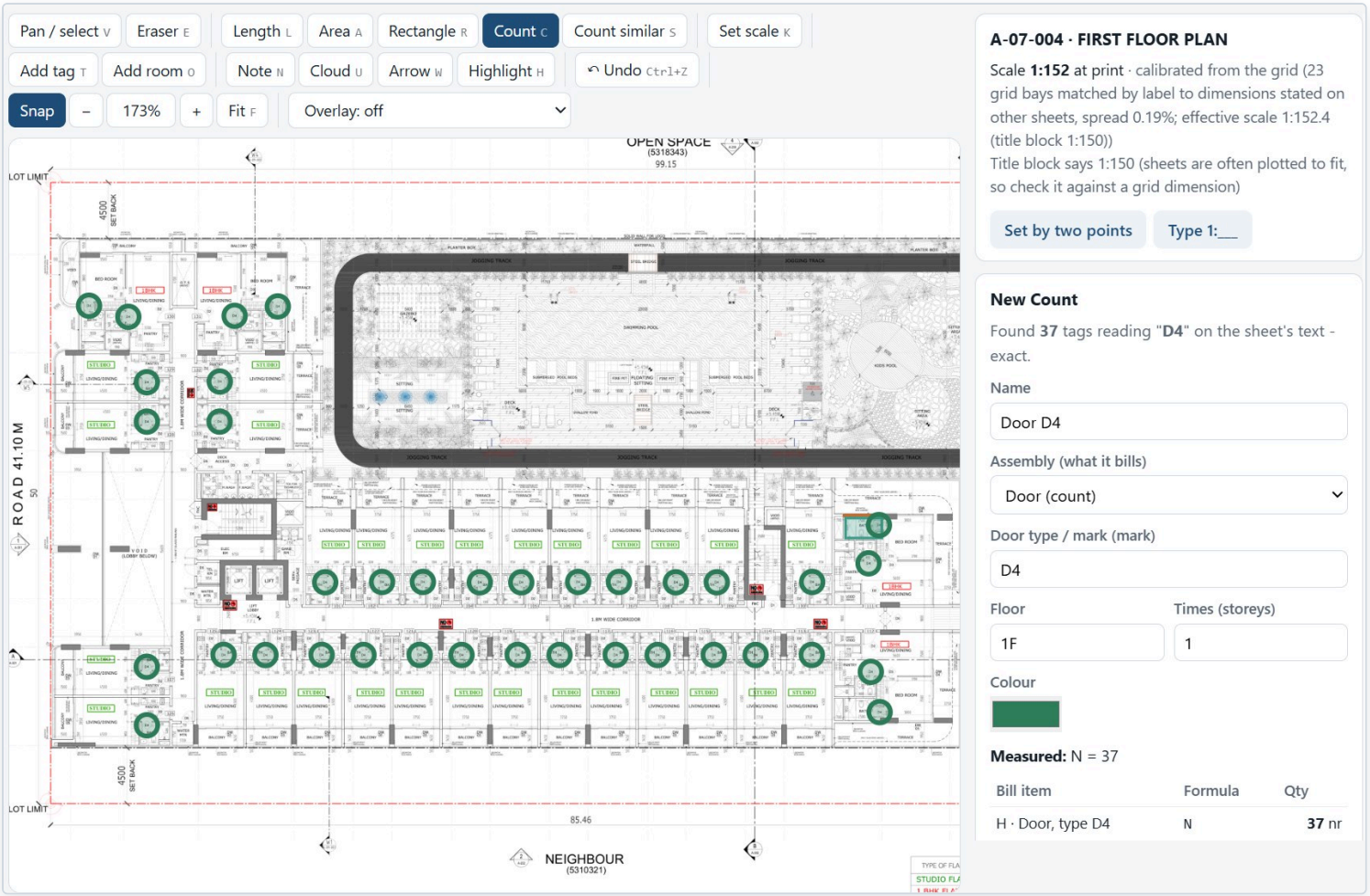
- A 3.10 m wall with *Block wall (both faces plastered)*: 11.00 m² blockwork (L×H) and 22.00 m² plaster (2×L×H)
- A rectangle over a bath with *Room finishes*: floor, skirting, walls and ceiling

WORKING

- Double-click to start and finish; Shift keeps a line straight; Backspace takes back a point
- Scroll zooms round the pointer; the view scrolls at the edges while measuring
- Times (storeys) for typical floors; deductions for voids; a slope for anything measured on plan over one (page 23)
- Notes, clouds, arrows and highlights, burnt into a marked-up PDF on export
- Eraser, and Undo for every change

Count similar: box one, find them all

Box one symbol or tag and every match on the sheet is ringed for you to check. A tag with real text is found by its text, which is exact; a graphic symbol is found by image matching in four orientations.



Count similar on the first floor plan: every D-4 tag ringed.

EXAMPLE

- One D-4 tag boxed on the first floor plan: **37 found**, the same number the automatic take-off counted there
- The Door assembly and the mark are filled in for you

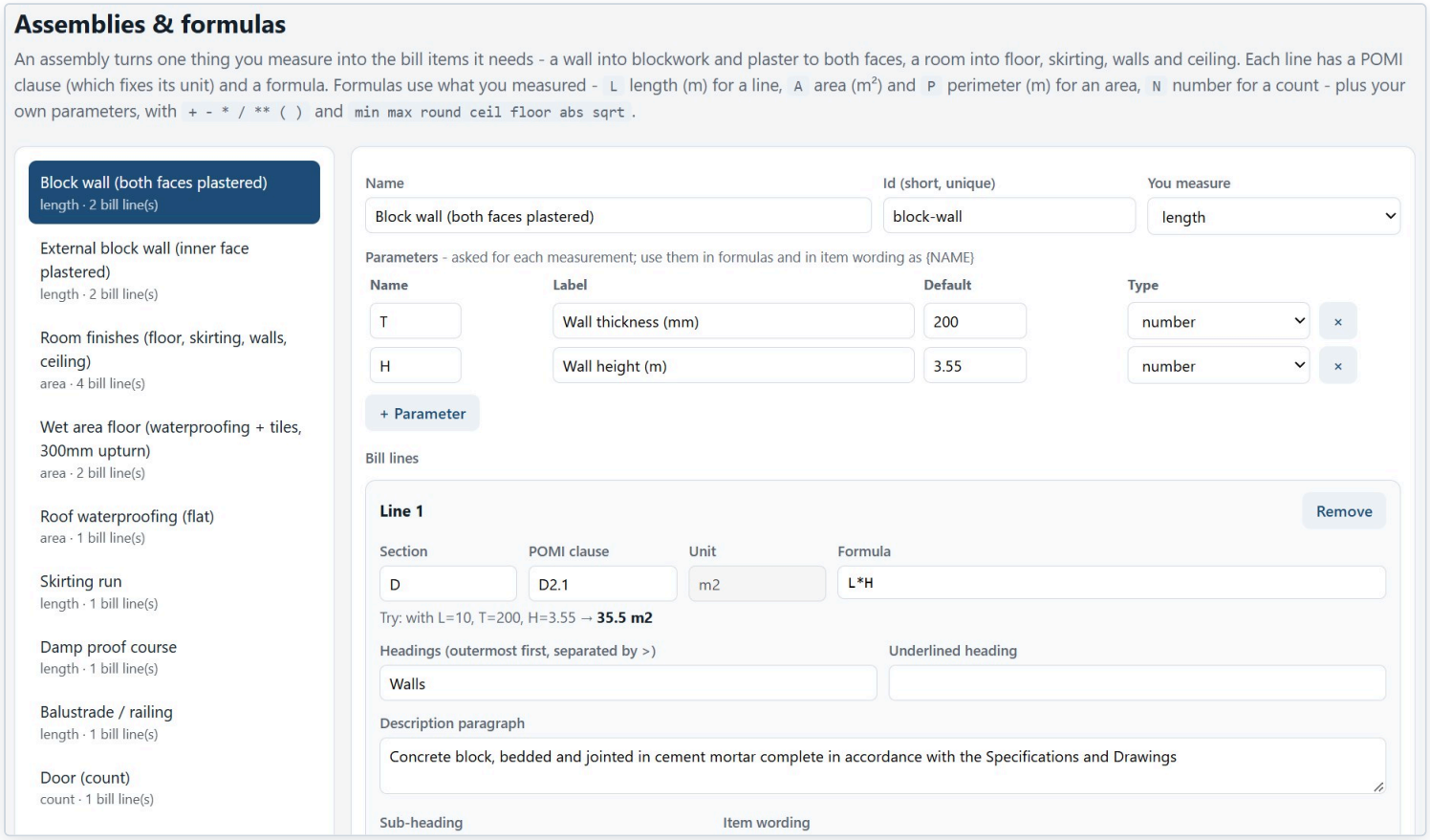
CHECKING

- Click a ring to remove it, click to add a missed one
- For image matches, a strictness slider re-runs the search

MEASURING BY HAND

Your own assemblies and formulas

An assembly turns one thing you measure into the bill items it needs. Each line has a POMI clause, which fixes its unit, its headings and wording, and a formula.



The Formulas tab, editing the block-wall assembly.

FORMULAS

- Use what you measured: `L` length, `A` area, `P` perimeter, `N` count
- Plus your own parameters, e.g. wall height `H` and thickness `T`
- Arithmetic only (`+ - * / **`, `min max round ceil floor abs sqrt`); never executed as code
- *Try* shows the result with sample values as you type

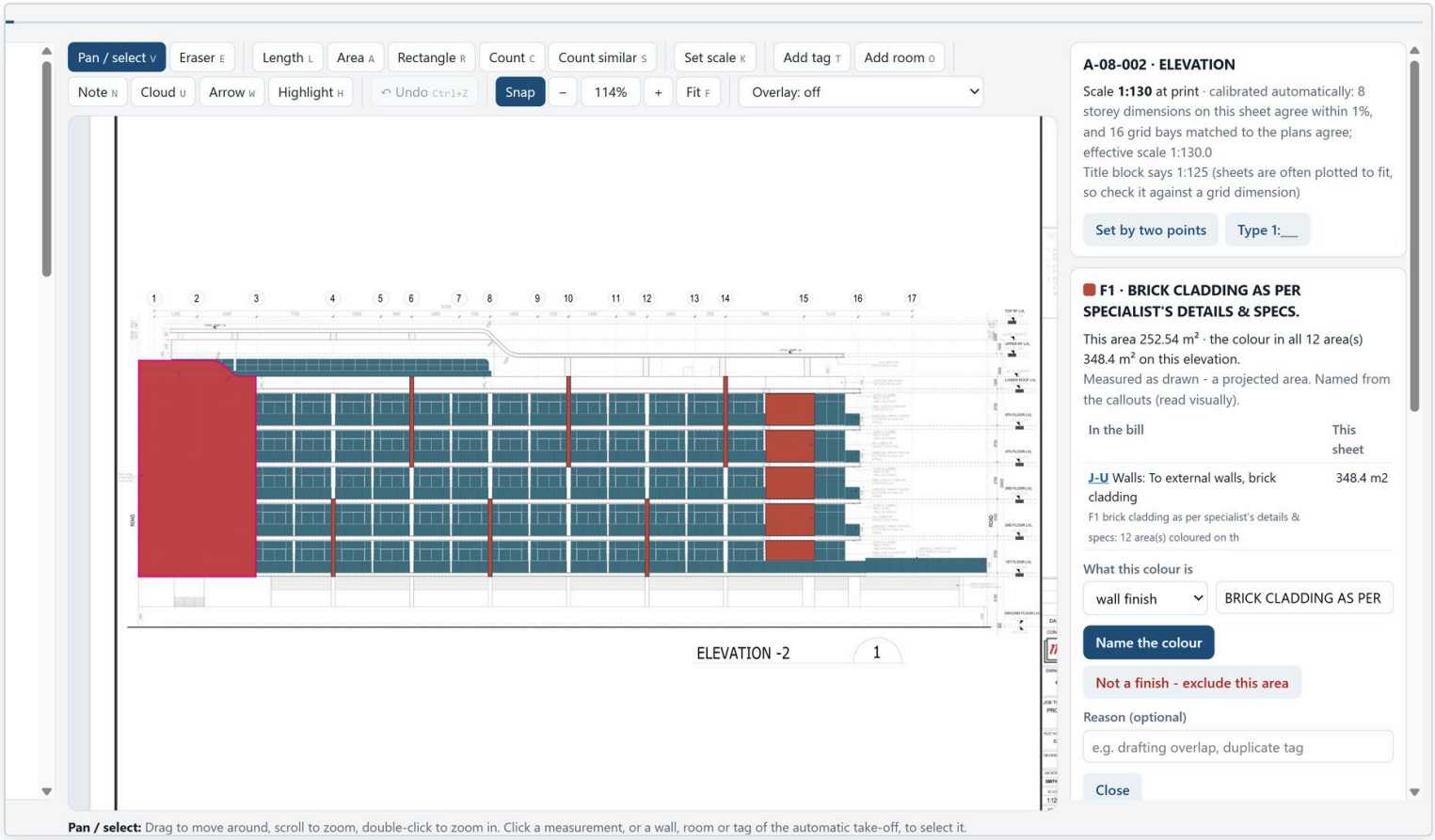
LIBRARY

- Block walls, room finishes, wet-area floors, roof waterproofing, skirting, DPC, balustrades, doors, windows

ELEVATIONS

External finishes measured by colour

The elevations show their finishes as colour. Each coloured area is measured at the elevation's own scale, the callouts say which finish each colour is, and wall finishes are billed as external finishes, one line per elevation.



A-08-002 in Measure: one brick-cladding area selected, its share of F1 and the bill line it feeds; the scale note above.

SCALE

- From the elevation's own storey dimension chain: every floor-to-floor figure against its tick marks, all within 1%
- Checked against the plans' grid: A-08-002, 8 storey dimensions and 16 grid bays agree; **1:130.0** (title block 1:125)
- Test set: 1:130.0, 1:130.0, 1:127.8 and 1:149.4

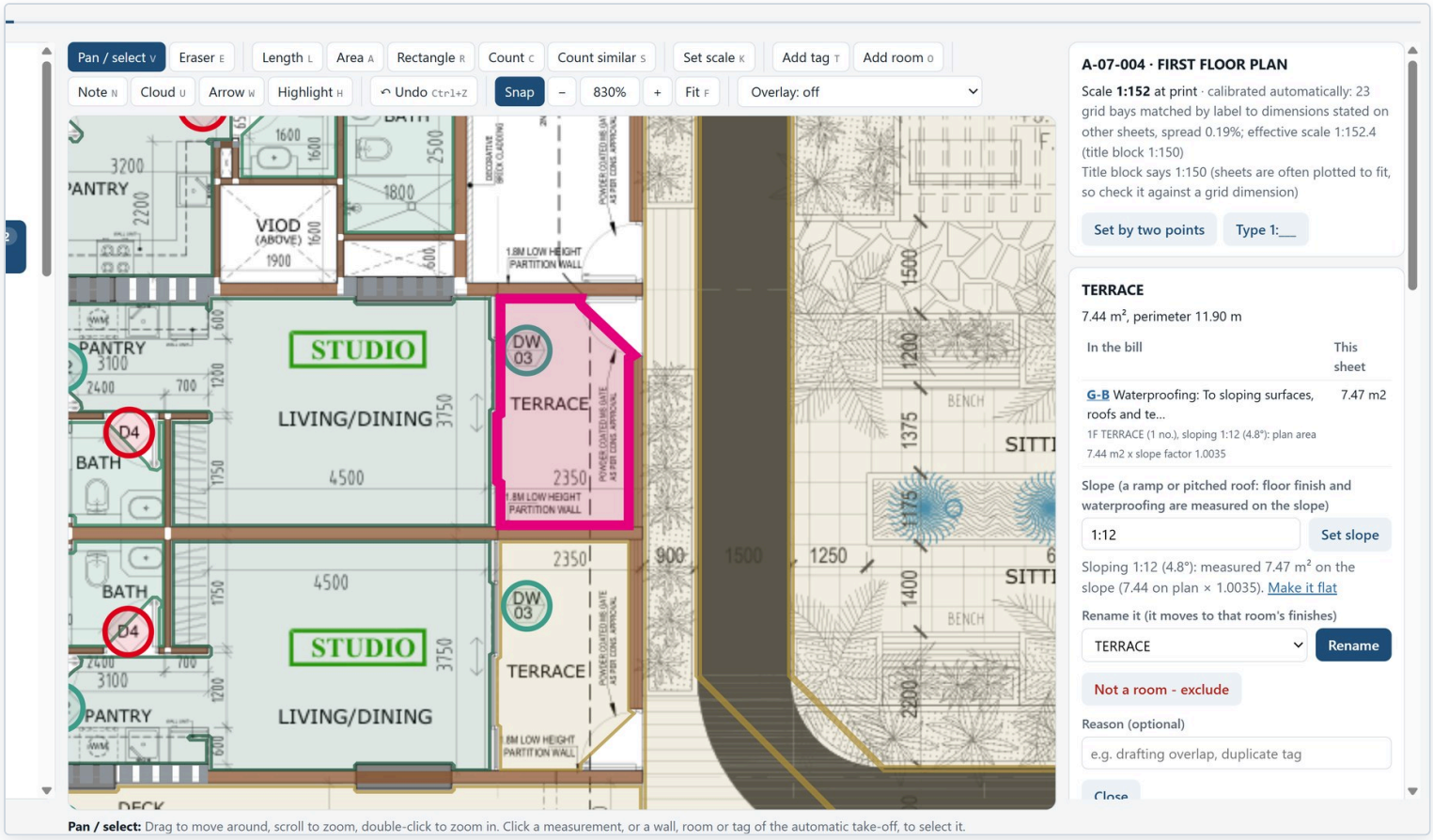
MEASURING AND BILLING

- One hue is one finish class; lines over a fill are closed over, specks and patches under 0.3 m² dropped
- The callouts are read visually against labelled colours; wall finishes go to J3.1 as *To external walls, <finish>*
- A-08-002: **F1 brick cladding 348.40 m²** (J-U); F2 glazing 862.43 m² is billed from the schedules, not here
- A colour no callout names is a gap, never billed (A-08-004, 87.3 m²); name it in Measure, or exclude an area

SLOPES

Measured on the slope, not on plan

Anything measured on plan over a slope is shorter than it is. Give it its slope as a pitch, a gradient or a percentage and it is measured on the slope, with its own sloping bill items.



A terrace on the first floor plan given a slope of 1:12, and the sloping waterproofing line it now feeds.

HOW A SLOPE IS WRITTEN

- `15` (degrees), `1:12` or `1 in 12`, `8%` ; the factor is 1 / cos(pitch)
- Flat, zero or anything steeper than a sensible slope is refused, not guessed

ROOMS FROM THE TAKE-OFF

- Floor finish and waterproofing measured on the slope as their own items (G2.1 sloping; J …, *sloping*)
- Walls, skirting and ceiling stay as on plan
- Example: a terrace, 7.44 m² on plan, at 1:12 → **7.47 m²** (× 1.0035) under **G-B** *To sloping surfaces*

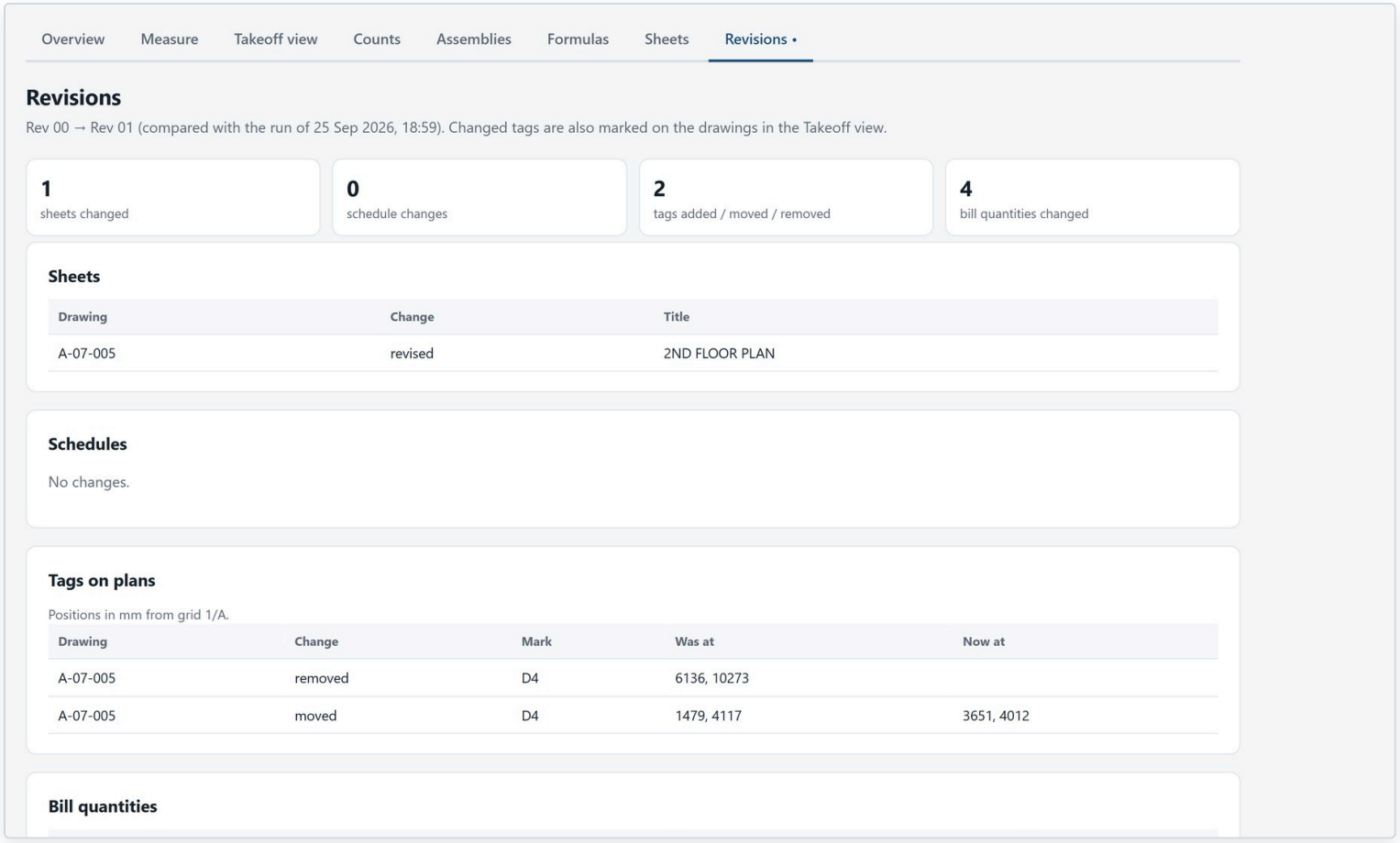
MEASURED BY HAND

- Length, Area and Rectangle take a Slope: `L` and `A` on the slope, `Lp` / `Ap` on plan, `F` the factor
- Assemblies *Roof covering (sloping)* and *Ramp / sloping floor finish*

REVISIONS AND EXPORT

What changed since the last issue

Each run is compared with the previous run of the same project: sheets revised, schedule changes, tags moved, added or removed, and bill quantities changed. The changes go to a Revision Changes workbook and are marked on the drawings.



The Revisions tab after the revision test (rev 00 → rev 01): 1 sheet revised, 2 tags moved or removed, 4 bill quantities changed.

IN THE REVISION TEST

- One D-4 tag moved 2 m and another deleted on the 2nd floor plan
- Found: the sheet revised, 1 tag moved, 1 removed; D-4 reopened; 4 quantities changed

CARRYING WORK FORWARD

- **Overlay** in Measure: a sheet over the same sheet from another run (grey same, red removed, green added)
- Manual measurements and corrections are brought into the new run; revised sheets are flagged
- On a revised sheet each corrected shape is found again where it was; any that cannot be placed are listed

HOW IT STAYS ACCURATE

The model reads; code computes

AI is used only to read what exists solely as graphics. Every number that reaches a workbook is arithmetic done in code on cited facts, and every run checks itself before it hands anything over.

Reading

- Text from the PDF text layer where it exists
- Vector geometry measured at a scale calibrated for each sheet from its grid, never the title block; each elevation from its own storey dimensions, checked against the plans' grid
- AI reads only schedule cells, dimension strings, level values and detail notes drawn as graphics
- Each reading is saved, stamped with the sheet's fingerprint and the prompt version; a changed drawing is read again

Measuring

- Every quantity carries its origin and its drawing
- Counts are made two ways; only agreeing counts are billed
- POMI rules applied in code: units, deductions, add-backs, required wording
- Nothing is guessed: what is unclear goes to the gaps list
- Hand corrections are measured again by the same rules

Checking

- Every bill quantity recomputed from the take-off's own dimensions and compared
- House template: column widths, 62-row pages, headings on every page, To Collection sums, item letters, fonts
- Checked on every run and every export
- Automated tests: unit checks, Section H acceptance (480 doors, 199 windows), revision diff, corrections, slopes and external finishes

Privacy

- The platform runs on your own computer. Runs are kept in `boq_arch\jobs\`. API keys are entered per run and not stored; only the graphic regions that need reading are sent to the AI provider.

SCOPE

What is measured today, and what is not yet

This version covers the main architectural sections. What it does not measure is listed in the gaps on every run, so nothing silently drops out of the bill.

Measured

- **D Masonry**: block walls by thickness (net length × storey height less slab), balcony partitions, openings (GP3.2)
- **G Thermal & moisture**: waterproofing to roofs, terraces, balconies and wet areas, flat or sloping; DPC under ground-floor walls
- **H Doors & windows**: one item per schedule mark, size from the schedule or elevations, counts reconciled
- **J Finishes**: floors, wall plaster, skirtings and ceilings by room class, on the slope where given one; external wall finishes by colour on the elevations

Not measured yet (listed in gaps)

- External finishes left uncoloured on the elevations (paint, stone, aluminium profiles and slats), and faces drawn faded
- Railings and balustrades
- Ironmongery (no hardware schedule in the set)
- Thresholds
- Roof build-up choice where several systems are detailed
- Anything on sheets missing from the set

Good to know

- Upload the complete set: a scale, storey height or finish may come from another sheet
- Sheets drawn as graphics need an AI key the first time (5 were left unread in the test run without one)
- DWG files need the free ODA File Converter
- Excluding a wall by hand does not re-flood the rooms; their areas stay as measured
- The bill is unpriced on purpose; no rate is ever invented